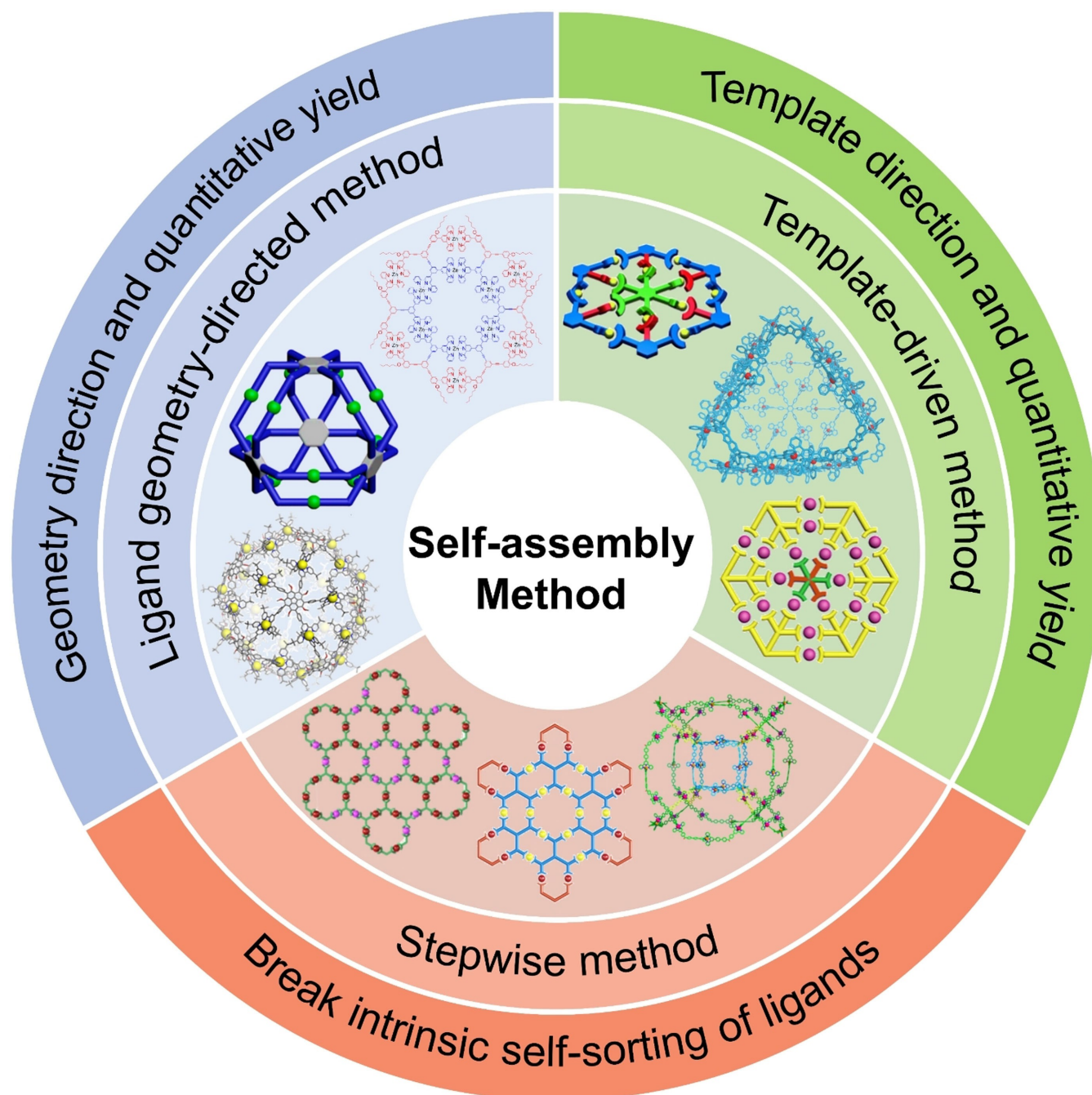


# Self-Assembly Methods for Recently Reported Discrete Supramolecular Structures Based on Terpyridine

Junjuan Shi and Ming Wang<sup>\*[a]</sup>



**Abstract:** In this Review, self-assembly methods of discrete metallo-supramolecules based on 2,2':6',2''-terpyridine (tpy) are comprehensively summarized. With the development of self-assembly, strategies for discrete 2D and 3D supramolecular architectures have boomed, including the geometry-directed method, template-driven method, and stepwise method. Ligand geometry-directed method mainly depends on the geometry of ligands (i.e., angle, geometric strain, and rigidity), and it is suitable for dual-component

systems, while the template-driven method can guide the self-assembly of predesigned supramolecules by the introduction of specific templates. Meanwhile, stepwise method, breaking the inherent self-sorting of ligands and reducing the probability of mismatch, is suitable for multicomponent systems to yield predesigned supramolecules. This review focuses on self-assembly methods and aims to provide a guideline for constructing supramolecular architectures using a suitable approach.

## 1. Introduction

Coordination-driven self-assembly, which renders high directionality as well as moderate bonding strength, is an appropriate approach to fabricate numerous discrete supramolecular structures with fascinating chemical and physical properties.<sup>[1–3]</sup> Numerous innovative studies on supramolecular self-assembly reported by Lehn,<sup>[4]</sup> Stang,<sup>[5]</sup> Stoddart,<sup>[6]</sup> Raymond,<sup>[7]</sup> Clever,<sup>[8]</sup> Fujita,<sup>[9]</sup> Leigh,<sup>[10]</sup> Nisthke,<sup>[11]</sup> Newkome,<sup>[12]</sup> and others<sup>[13]</sup> have demonstrated that coordination-driven self-assembly is a well-documented strategy for constructing a broad array of metallo-supramolecules with well-defined shapes, sizes, and geometries.

Among a wide variety of organic ligands, 2,2':6',2''-terpyridine (tpy) as a building block has attracted considerable attention in coordination-driven self-assembly<sup>[14]</sup> due to its remarkable binding ability to diverse transition metal ions with different binding strengths. The < tpy-M-tpy > motif exhibits a pseudo-octahedral coordination geometry, and owing to its electronic and steric versatility, it plays a prominent role in the construction of a series of self-assembled supramolecular structures. By employing terpyridine, a number of delicate two-dimensional (2D) and three-dimensional (3D) structures have been successfully prepared,<sup>[14]</sup> including triangles, hexagons, cubes, cuboctahedrons, and spheres, etc. With further study, well-defined sophisticated metallo-supramolecules with increasing sizes and complexity have become extremely challenging.

In recent years, predesigned supramolecular architectures have attracted considerable attention. For pursuing high-ordered metallo-supramolecules, it is crucial to develop synthetic and self-assembly methods. In the self-assembly process, preprogrammed multitopic ligands with tpy do not exhibit selectivity with ligands and different metal ions, and the competitive binding affinity in multicomponent systems hinders precise self-assembly. To deal with the problems, researchers have endeavored to develop several novel methods to control the self-assembly process to avoid unwanted products. Inspired by the structural complexity and diversity in nature, chemists have devoted their efforts to create intricate assemblies. Hence,

it is imperative to develop efficient self-assembly strategies to construct artificial supramolecules with immense intricacy.

In this Review, self-assembly methods for terpyridine-based metallo-supramolecules are focused. First, the advantages and disadvantages of each method are briefly introduced. Next, the reported approaches are classified. We are hopeful that this Review can serve as a guideline for the precise design of elegant metallo-supramolecular structures. According to different self-assembly systems, the three main approaches are as follows: (1) ligand geometry-directed method, (2) template-driven method, and (3) stepwise self-assembly method. In addition, representative examples and advantages of each method are reported. Some excellent reviews have provided further insights about the recent progress in the area of discrete metallo-supramolecular architectures based on terpyridine<sup>[14–15]</sup> as well as applications about the discrete structures, polymers, and other materials.<sup>[16–17]</sup>

## 2. Ligand geometry-directed method

For pursuing structural complexity and high ordering of metallo-supramolecules, chemists have developed various methods to reduce the possibility of misconnection in a complicated system. Among these methods, ligand geometry-directed assembly is typically employed to construct metallo-supramolecules due to the affinity for systems with labile transition metals, such as  $\text{Zn}^{2+}$  and  $\text{Cd}^{2+}$ . All components go through exchange processes of possible intermediate states and finally reach the most thermodynamically stable states. Of course, nonlabile metal ions (e.g.,  $\text{Ru}^{2+}$  and  $\text{Fe}^{2+}$ ) are also utilized to construct metallo-supramolecules with suitable ligands. Specific supramolecules are mainly determined by the geometry of building blocks, such as steric hindrance, angle of ligands, and length of arms.

For precisely controlling the self-assembly behavior and constructing well-defined and complex metallo-supramolecular architectures, Li et al. have developed a new method based on the conventional self-assembly and designed tritopic and tetra-topic 2,2':6',2''-terpyridine building blocks **1** and **3** with an angle of  $120^\circ$  between two ligating moieties to afford hexagons **2** and **4** (Figure 1),<sup>[18]</sup> and overcame challenges of a mixture of metallomacrocycles.<sup>[19–20]</sup> During the progress of constructing the supramolecules, the concept “density of coordination sites” (DOCS) was proposed, which played a key role in guiding

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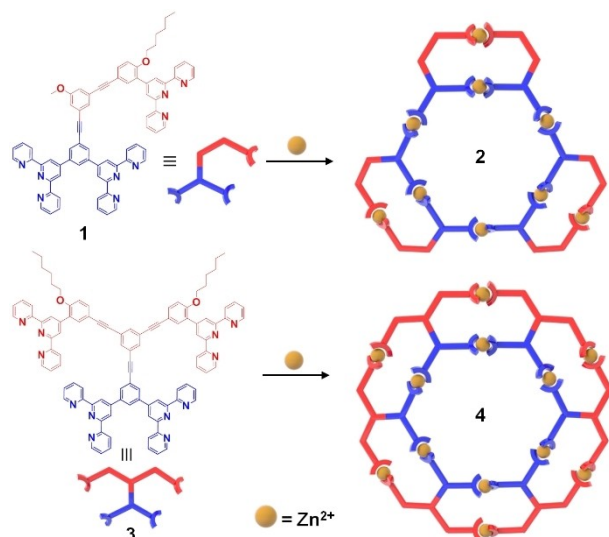


Figure 1. Structures of hexagon wreaths 2 and 4.

Junjuan Shi received her BS in 2016 from Changchun University of Technology, and she completed her Ph.D. in 2021 under the guidance of Professor Ming Wang at Jilin University (China) focusing on discrete 2D terpyridine-based supramolecules. During 2018–2020, she joined the group of Prof. Xiaopeng Li at the University of South Florida (US) as a joint training of doctoral student. Currently, she is a Postdoctoral Researcher at Jilin University under the “Dingxin Scholar Program.” Her research interests include the construction and function of 2D supramolecular architectures.



Ming Wang received his BS in 2004 from Jilin University and Ph.D. in 2010 from the Changchun Institute of Applied Chemistry, Chinese Academy of Sciences. After receiving his Ph.D., he joined Professor Pooi See Lee's group in Nanyang Technological University (Singapore) as a Postdoctoral Researcher. In 2013, he joined the group of Professor Xiaopeng Li as a Postdoctoral Researcher in Texas State University (US). His research focuses on the design and synthesis of 2D and 3D metallo-supramolecules based on terpyridine. In 2016, he was appointed as a Professor in the College of Chemistry, Jilin University. His research interests include the construction of metallo-supramolecular architectures, self-assembly based on dissymmetrical ligands, and functional design of metallo-supramolecules.

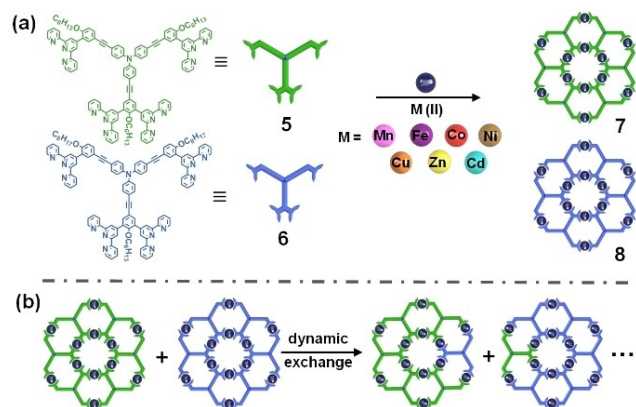


precise self-assemblies by providing a high geometric direction, especially in thermodynamically controlled systems. Owing to this method, a series of high-generation metallo-supramolecules, such as concentric hexagons,<sup>[21]</sup> Kandinsky circles,<sup>[22]</sup> and rosettes,<sup>[23]</sup> have been designed.

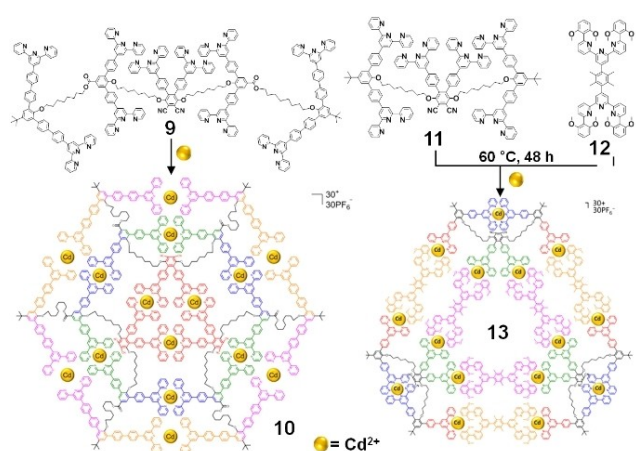
For further broadening the diversity of metallo-supramolecules, seven bivalent transition metal ions (M = Mn, Fe, Co, Ni, Cu, Zn, and Cd, respectively) were used in the self-assembly of supramolecules to explore the dynamics of self-assembly. A series of hexameric metallo-supramolecules 7 and 8 were prepared by tetratopic terpyridine ligands with different alkyl chains and seven metal ions (Figure 2).<sup>[24]</sup> Structural stabilities of the supramolecules in the gas phase were investigated by gradient tandem mass spectrometry, which decreased in the order of Ni > Co > Zn > Fe > Cu > Cd > Mn. Kinetic aspects of the dynamic ligand-exchange process between two discrete supramolecules in solution were employed to explore structural stability using ESI-MS, indicating that ligand exchange occurs at 120 °C.

Except for firm linkers between two layers in the ligand, the adjustable flexible linkers between ring-in-ring ligands are introduced to precisely construct supramolecules. Chan has reported well-defined ring-in-ring structures with multivalent ligands with flexible linkers.<sup>[25]</sup> The geometrical constraint from flexible linkers is the key factor for controlling self-assembly compositions and the stability of ring-in-ring architectures from the results of DOSY and ESI-MS. Ligand 9 with C8 linkers was successfully assembled into 10 without other structures (Figure 3). To further study the formation mechanism for ring-in-ring assembly, an intramolecular complex was first prepared by a titration experiment before the final ring-in-ring structure was constructed. With the discovery of complementary terpyridine pairing, a multicomponent ditrion 13 was constructed by the single-pot reaction of ditopic ligand 12 modified by 2,6-dimethoxyphenyl groups at the 6,6'-positions, ligand 15, and Cd(NO<sub>3</sub>)<sub>2</sub> in a 1:1:2 ratio.<sup>[26]</sup> In the next few years, Chan has also constructed a series of fascinating self-assembled macrocycles, cages, and nanocapsules by this method.<sup>[27–30]</sup>

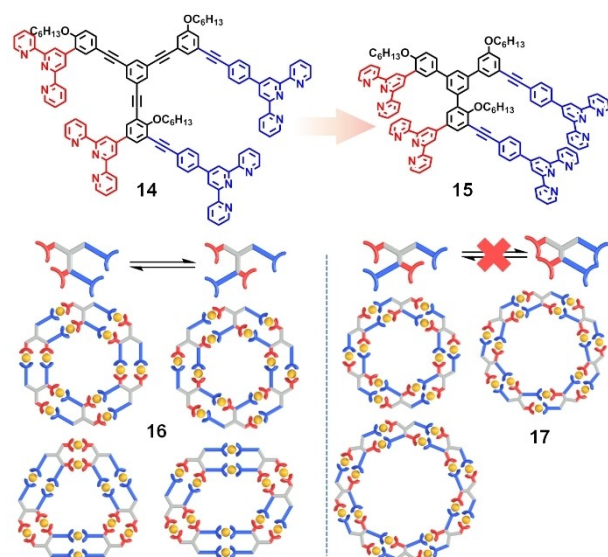
All of the above architectures have been constructed with symmetrical ligands. In complex systems for precise self-assembly of well-defined and functional organisms created by nature, the widespread dissymmetric structures play a key role. However, dissymmetric elements merely exist in artificial supramolecules, due to tedious synthetic routes of low symmetrical ligands, presence of isomers with similar physical properties, and characterization of complicated structures. Recently, novel supramolecular systems with dissymmetric terpyridine building blocks 14 and 15 have been developed (Figure 4).<sup>[31]</sup> Complexes 16 assembled by loose dissymmetric building blocks 14 contained isomers and could not be distinguished by NMR, ESI-MS, and TWIM-MS. STM was utilized to visually analyze the self-assembled isomers at the sub-molecule level. From STM data analysis and DFT calculations, four isomers existed in complexes 16, corresponding to four coordination modes. The coordination mode was controlled via the introduction of steric constraints, and there was only one coordination mode in complexes 17 with compact ligand 15.



**Figure 2.** (a) Self-assembly of hexameric fractals through 5 and 6 with seven transition metal ions and (b) dynamic ligand exchange between two fractals and the same metal ion.



**Figure 3.** Self-assembly of ring-in-ring structure 10 and ditrign 13.

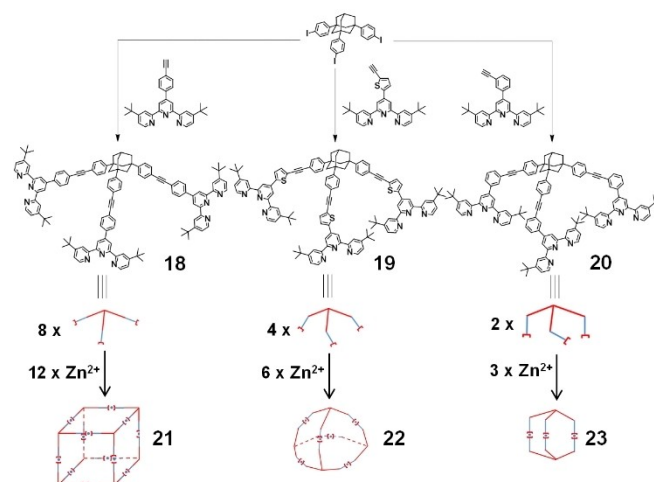


**Figure 4.** Self-assembly of dissymmetric ligands 14 and 15 for the construction of complexes 16 and 17.

It is well known that artificially obtaining 3D supramolecular architectures based on terpyridine is challenging. The tpy-M-tpy subunit is generally connected at 180°; hence, it is limited to be corner directing units.<sup>[32]</sup> Therefore, researchers have focused on the design of appropriate tpy-based ligands with directing units to obtain an array of three-dimensional metallo-supramolecular architectures.

In 2014, Li and coauthors have designed a series of 3D metallo-supramolecular architectures with tritopic terpyridine ligands with adamantane as corner directing units (Figure 5).<sup>[32,33]</sup> Cube 21, tetrahedron 22, and metalocage 23 were obtained by changing the angles of the linkers between adamantane and tpy, which played a critical role in the self-assembly process. ESI-MS, TWIM-MS, and NMR were employed to characterize structural details and molecular composition. In addition, the clear geometry direction of the different ligands resulted in self-sorting behaviors of these 3D structures, which were investigated by mixing two ligands of 18–20, and the self-sorting behavior was solely based on the angles precoded within the arm of tritopic ligands. Kinetic investigation of a mixture of supramolecular structures 21 and 22 revealed that the signals of the two 3D structures disappear after 7 days from ESI-MS.

In a balance of the directionality of ligands and flexible linkers, Newkome and coauthors have designed and synthesized an expanded tetrahedral structure 25 with four independent triangular surfaces with four flexible vertices using tribenzo-27-crown-9 ether under thermodynamic control (Figure 6).<sup>[34]</sup> Building block 24, a crown ether, is used as the core with three bis-ligands juxtaposed at 60°. Owing to the flexibility of crown ethers and the stability of triangles, the tetrahedron was constructed successfully. The internal volume of cationic tetrahedron 25 was an excellent candidate to encapsulate the large tetrahedral anion tetrakis(perfluorophenyl)borate (BARF). Following the design concept, two metallo-prisms 27 and 28 with dibenzo[24]crown-8 units were designed using tetratopic terpyridine ligands and  $\text{Cd}^{2+}$  or  $\text{Zn}^{2+}$  in quantitative yields (Figure 7).<sup>[35]</sup> As a host, metalloprism 27 with three crown



**Figure 5.** Synthetic route of ligands and complexes 21–23.

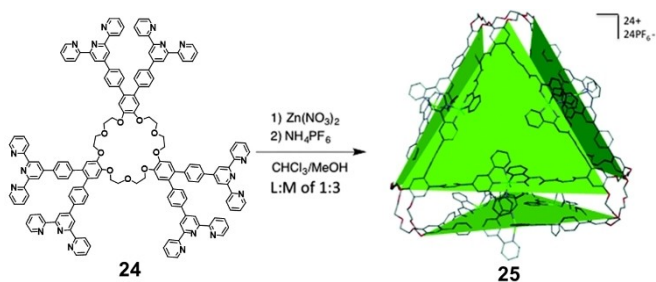


Figure 6. Self-assembly of tetrahedron 25 with ligand 24 and  $\text{Zn}^{2+}$ .

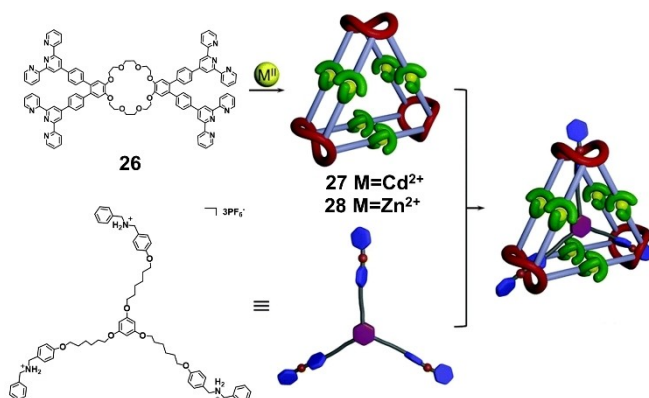


Figure 7. Self-assembly of metallo-supramolecules 27 and 28 and host-guest complexes.

vertices could occupy the guest tris(dibenzylammonium)· $3\text{PF}_6$  salt (TriG) in a 1:1 ratio by using time-dependent  $^1\text{H}$  NMR experiments.

Inspired by the construction of 2D concentric hexagons based pyridinium salts and terpyridine,<sup>[21,22]</sup> Li has designed and assembled hexagonal prism 30 using pentatopic ligand 29 with different alkyl chains and  $\text{Cd}^{2+}$  by the geometry-directed method (Figure 8).<sup>[36]</sup> The self-assembly mechanism of these prism structures was further studied and monitored by ESI-MS, revealing an intermediate state for the assembly of concentric hexagons with six metal-free terpyridines. The antimicrobial activity of hexagon prisms was explored, and the antimicrobial material exhibited high selectivity and binding affinity for Gram-positive bacteria.

With the modification of functional tetraphenylethylene (TPE), two supramolecular cages 33 and 34 were assembled by

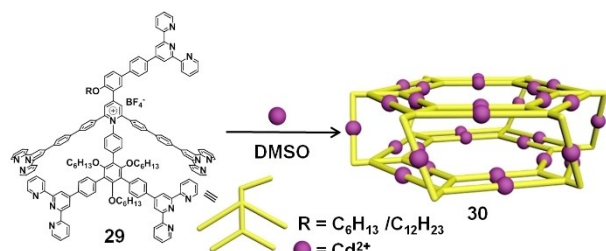


Figure 8. (a) Self-assembly of hexagonal prisms 30.

tetrapotic TPE-decorated terpyridine (tpy) ligands 52 and 53, respectively, with  $\text{Zn}^{2+}$  (Figure 9).<sup>[37]</sup> Compared with 31, extra alkyne was introduced into ligand 32 to increase the geometric constraint. Furthermore, emission behaviors of the ligands and cages in the solid state under external pressure were examined. With the gradual increase in external pressure, the luminescence of 34 increased initially, which was related to the further restriction of rotation. Subsequently, luminescence quenching was observed at 6.32 GPa, which was caused by tightly packed supramolecules. The subsequent release of pressure led to emission recovery.

For further exploiting supramolecular functions, Wang and coauthors have reported and constructed a series of twist supramolecules 38–40 that were assembled using multitopic terpyridine ligands via ortho modification (Figure 10).<sup>[38]</sup> At high concentrations, twisted dimers 38 exhibited mechanically interlocked forms as [2]catenane and [3]catenane, which were observed with the increase in concentration. With the increase in the space constraints, the interlocking phenomenon was not observed for the system of 40 assembled by hexatopic ligand

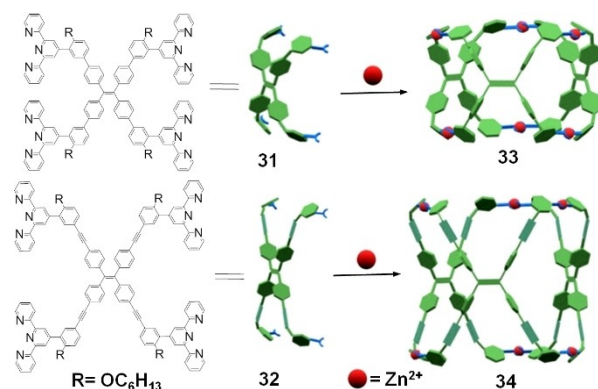


Figure 9. Self-assembly of complexes 33 and 34.

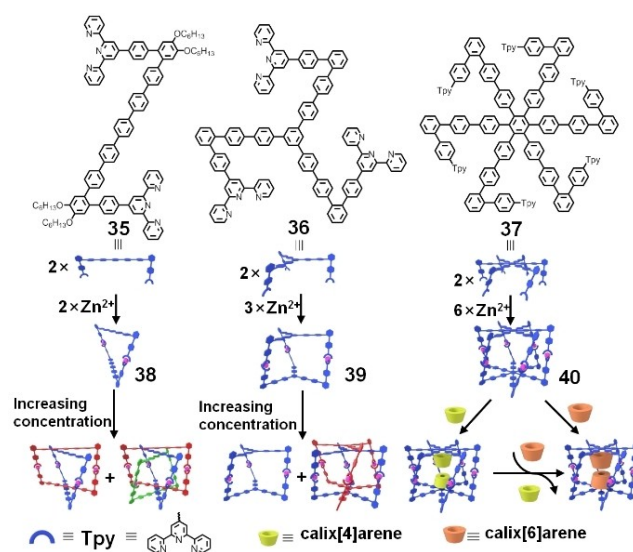


Figure 10. Self-assembly of discrete twisted dimer supramolecular architectures 38–40, mechanically interlocked structures, and guest recognition.

37 and  $\text{Zn}^{2+}$ . Interestingly, metallo-cage **40** encapsulated guest calix[4]arene and calix[6]arene, and calix[6]arene could exchange the guest calix[4]arene from host metallo-supramolecule **40**, suggesting that cage **40** exhibits a high affinity for calix[6]arene.

With the introduction of special linkers, Newkome has reported a classic Archimedean solid-cuboctahedrons **42** and **43** with X-shaped tetradentate terpyridinyl ligands **41** by single-step self-assembly (Figure 11).<sup>[39]</sup> The building block with two desired angles (i.e.,  $60^\circ$  and  $90^\circ$ , respectively) can be utilized to construct triangles and squares in supramolecular structures with  $\text{Cd}^{2+}$  or  $\text{Zn}^{2+}$ . The crystal structure of metallo-supramolecule **42** was characterized by synchrotron X-ray analysis. Cuboctahedron **43** could be converted into an octahedron by dilution. Except that, the stability of the assembled structure was also affected by counterions' size.

More recently, one of Johnson solids-triangular orthobicupola **45**, with a higher similarity to Archimedes polyhedron-cuboctahedron, was constructed with multivalent terpyridine ligand **44** and  $\text{Zn}^{2+}$  in DMF (Figure 12).<sup>[40]</sup> The structural composition was confirmed by ESI-MS. The crystal data revealed that the structure of triangular orthobicupola **45** with eight terpyridine units is unambiguously confirmed. When the reaction solvent DMF was replaced by methanol, another supramolecular pentagonal prism **46** was obtained by self-assembly with the same components, and structural compositions were established by ESI-MS. Supramolecular conversion by changing the solvent can provide a method to encapsulate and release guest molecules.

Inspired by the aesthetically pleasing structures of spherical virus capsids, a discrete tetrameric metallo-nanosphere **48**

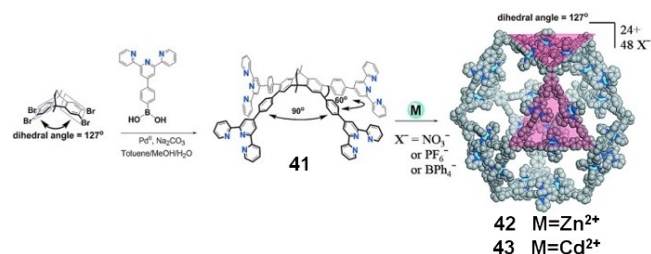


Figure 11. Self-assembly of metallo-cuboctahedrons **42** and **43** with  $\text{Zn}^{2+}$  and  $\text{Cd}^{2+}$ .

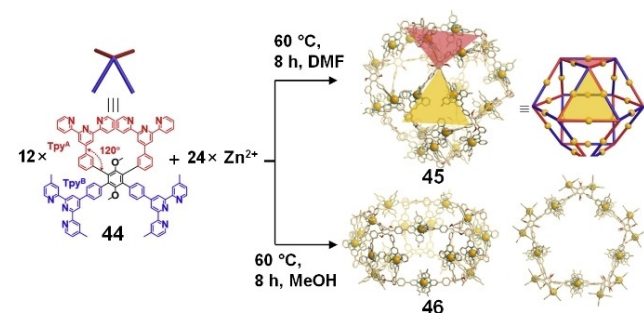


Figure 12. Self-assembly of triangular orthobicupola **45** and pentagonal prism **46**.

comprised a homologous-branched hexatripyridine ligand **47** and  $\text{Zn}^{2+}$  in nearly quantitative yield via one-step assembly (Figure 13),<sup>[41]</sup> according to the DOCS concept. The metallo-nanosphere exhibited four quasi-triangular and rhombus-like facets. For a stable metallo-capsule, the proper geometry and orientation required for the tpy-M(II)-tpy connectivity are necessary. Host-guest chemistry about nanospheres (internal volume is about  $9800 \text{ \AA}^3$ ) with coronene was studied by NMR and ESI-MS. More interestingly, stable metallo-sphere **48** was found to undergo self-assembly into berry-type structures in a mixed polar solvent.

In addition, to chemically mimic spherical virus capsids, a metal-organic capsid **50**, which was self-assembled with 12 bowl-shaped pentatopic corannulene-based skeletons **49** with a suitable curvature and 30  $\text{Cd}^{2+}$  linkers, was proposed by the Chan group (Figure 14).<sup>[42]</sup> The design concept of the ligand appeared to be straightforward, but the synthesis of bowl-shaped pentatopic tectons was extremely challenging. To increase solubility, the five methyl ethers of the precursor with 1,3,5,7,9-pentamethoxy-2,4,6,8,10-pentabromocorannulene and *t*-butyl groups at the terpyridyl 4,4''-positions were introduced in ligand **49**. Metal ion  $\text{Cd}^{2+}$  was selected for assembly with the < tpy-M-tpy > motifs with appropriate thermodynamic stability and kinetic flexibility. The external diameter of sphere **50** is about 6 nm, demonstrating potential to encapsulate hosts, recognize molecules, and so on.

Except for the unmodified terpyridine, Wang group has designed novel sterically congested ligands for geometrically oriented supramolecular self-assembly (Figure 15).<sup>[43]</sup> Ditopic ligand **51**, which exhibits steric constraints via the introduction of an unsymmetrical terpyridine subunit, can afford rhombus

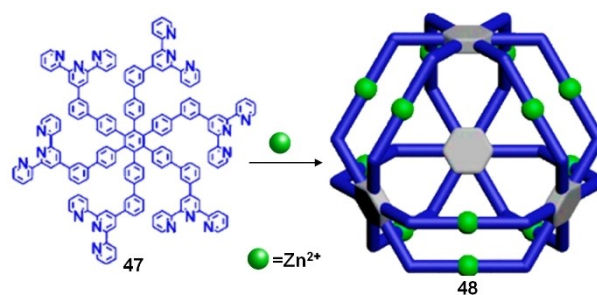


Figure 13. Self-assembly of a discrete spherical metallo-organic capsid **48**.

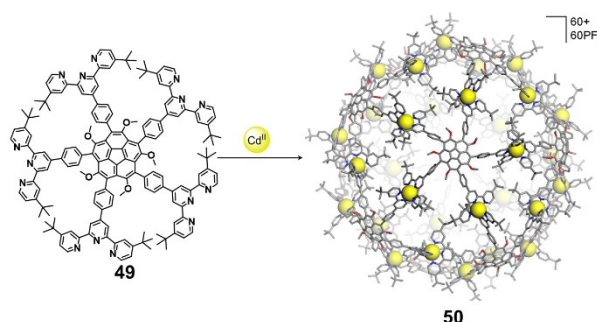


Figure 14. Self-assembly of ligand **49** with  $\text{Cd}^{2+}$ .

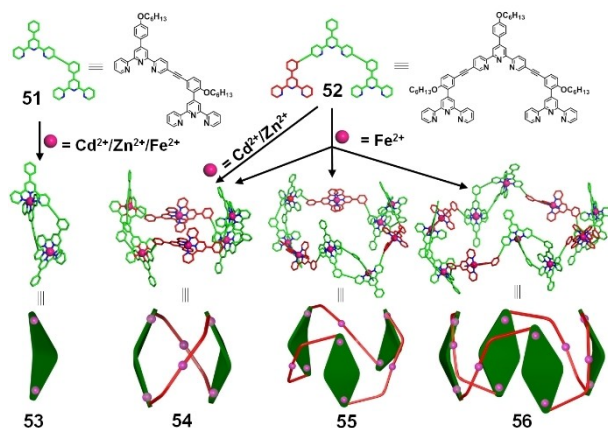


Figure 15. Self-assembly of ligands 51 and 52 with  $\text{Cd}^{2+}$ ,  $\text{Zn}^{2+}$ , and  $\text{Fe}^{2+}$ .

dimers **53** via assembly with  $\text{Cd}^{2+}$ ,  $\text{Zn}^{2+}$ , and  $\text{Fe}^{2+}$ . Tritopic ligand **52** and metal ions of  $\text{Cd}^{2+}$  and  $\text{Zn}^{2+}$  were assembled to afford tetramers **54**, but a mixture of tetramer, hexamer, and octamer was obtained via assembly with  $\text{Fe}^{2+}$  and controlled by kinetics and thermodynamics. The assembly mechanism was explored by using a mixture of **52** and  $\text{Cd}^{2+}$  in a stoichiometric ratio of 1:1, and the ESI-MS revealed that dimer intermediates are the key to ultimate supramolecular architectures.

### 3. Template-driven method

In multi-component metallo-supramolecular systems based on terpyridine, multi-structures with a similar thermodynamic equilibrium would be generated without ligand selectivity. It is difficult to control the geometry and connectivity and prevent the multi-component mismatch. With continuous efforts, researchers have developed a novel method-template-driven assembly. With the presence of templates, multicomponent self-assembly with directionality can be precisely controlled, and the template ligands not only serve as a coordination point with metal ions, but play a role in directing self-assembly to obtain well-defined architectures. Among numerous organic ligands, hexakis(terpyridine) **58** with  $D_{6h}$  symmetry is a "star molecule", which is selected as a template to guide the construction of discrete structures without oligomers or polymers.

As the first example of the template-driven method, Newkome has constructed shape-persistent 2D supramolecular spoked wheels **60** and **61** with a T-shaped ligand **59**, hexatopic ligand **58**, and metal ions ( $\text{Cd}^{2+}$  or  $\text{Zn}^{2+}$ ) by the template-driven method (Figure 16).<sup>[44]</sup> Hexatopic ligand **58** served not only as a template for the spoked wheel but also as a vertex for the smallest subunit-triangle. Based on the study, a three-dimensional spoked wheel **62** was successfully synthesized by a precise stoichiometric ratio (6:2:12) of **57**, **59**, and  $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$  in quantitative yield.<sup>[45]</sup> The backbone in the framework included two parts: the central spokes **57** and outer rims **59**. Ligand **57** exhibited two roles: the spoke and the

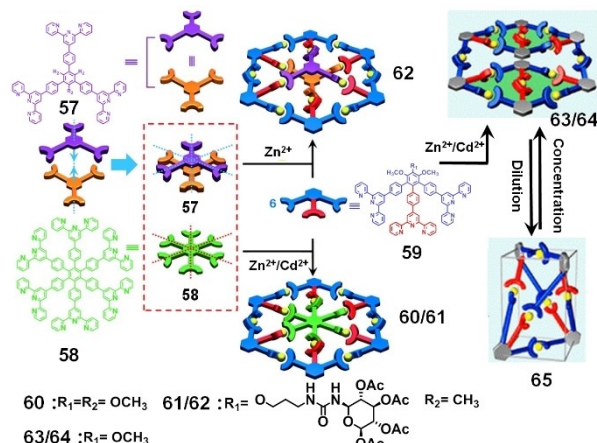


Figure 16. Synthesis of supramolecular 2D bicycle wheels **60** and **61**, 3D spoked wheel **62**, bis-rhomboidal shaped 3D molecular wheels **63** and **64**, and supramolecular interconversions between **64** and **65**.

template for self-assembly. Following the successful construction of spoked wheels, Newkome has designed a new bis-rhomboidal shaped, three-dimensional molecular wheel **63** by tritopic ligand **59** and  $\text{Zn}^{2+}$ , and the ligand **59** served as rims and internal hubs.<sup>[46]</sup> The dynamic conversions between spoked wheel **64** and tetrahedron **65** were examined by changing the concentration.<sup>[47]</sup>

Inspired by the above pioneering discoveries, Wang group has reported the self-assembly of double- and triple-decker spoked wheel supramolecular architectures **69–71** and **75–77** based on the  $\langle \text{tpy}-\text{Cd}^{2+}-\text{tpy} \rangle$  connectivity (Figure 17).<sup>[48]</sup> Multiple spoked wheels comprised multivalent terpyridine ligands (**66–68** and **72–74**) with different connecting lengths, hexatopic ligand **58** as the template, and  $\text{Cd}^{2+}$ . Recently, a fascinating

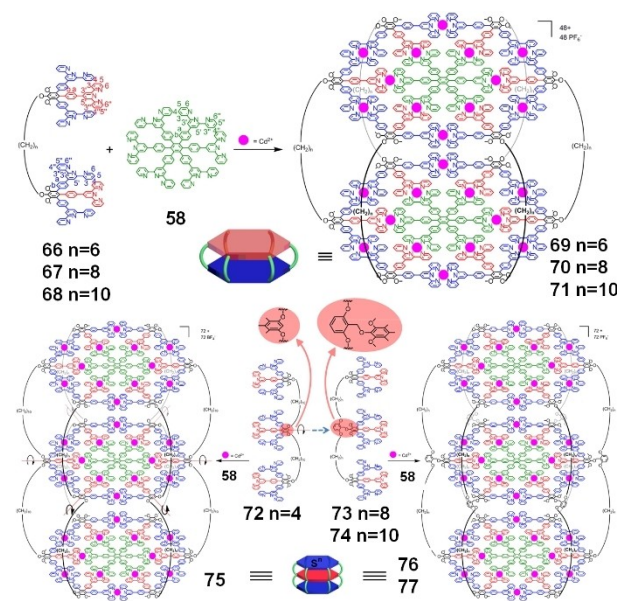


Figure 17. Self-assembly of double-decker hexagon **69–71** and triple-decker hexagon **75–77**.

truncated metallo-tetrahedron **79** with a molecule weight of up to 70 kDa was assembled by two organic ligands (**78** and **58**) and metal ions  $\text{Cd}^{2+}$  in a precise stoichiometric ratio of 12:4:48, comprising 64 components (Figure 18).<sup>[49]</sup> As vertexes of tetrahedron **79**, ligand **78** comprised two 2,8-position substituted trispyridines bridged by dibenzo[b,d]furan. The multicharged metallo-supramolecule could further assemble into highly ordered nano-aggregates with aggregation-induced emission enhancement and tunable emission color in mixed solvent systems.

To push the complexity of supramolecules, Li group has reported the one-pot synthesis of sophisticated 2D spiderwebs **82** and **83** with four components by the template-driven method (Figure 19).<sup>[50]</sup> The irreversible condensation of pyrylium salt (ligand **80**) and primary amine (ligand **81**) was successfully combined with multitopic terpyridine-based ligands **58** or **57** as templates. This new approach improved the synthetic efficiency of complex supramolecular architectures.

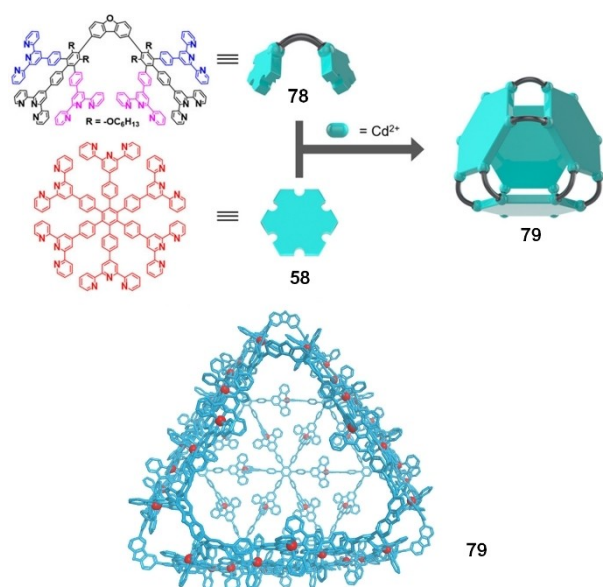


Figure 18. Self-assembly and energy-minimized structures of complex **79**.

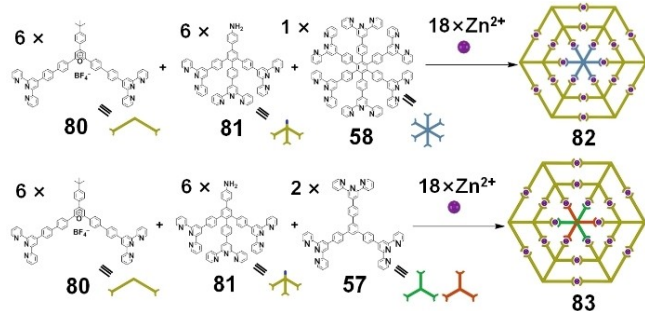


Figure 19. One-pot preparation of **82** and **83** via template-driven self-assembly.

## 4. Stepwise self-assembly method

In more complicate supramolecular system, it is incapable to satisfy the constructions of predesigned metallo-supramolecules by the one-pot method. In the pursuit of precisely controlling the construction of metallo-supramolecules, the stepwise method is crucial to design supramolecules with different ligands, break the intrinsic self-sorting property or chaotic coordination without selectivity, and prevent producing competitive products. The motifs  $\langle \text{tpy-M-tpy} \rangle$  with  $\text{Ru}^{2+}$  or  $\text{Fe}^{2+}$  were employed to link precursors to construct metallo-ligands by multi-step reactions. Finally, the supramolecules were assembled by metallo-ligands and metal ions. In recent years, stepwise methods also have successfully drawn extensive attention due to diverse and intricate supramolecular structures.

Based on the construction of the first-generation Sierpinski triangle by the geometry-directed method,<sup>[51]</sup> high-generation triangles **86** and **88** were assembled with  $\text{Ru}^{2+}$  by the stepwise method (Figure 20).<sup>[52]</sup> If ligands V and K were used in a precise stoichiometric ratio of  $\text{V/K/Zn} = 3:3:9$ , a mixture of triangle and **86** was obtained by the one-pot method. To avoid the formation of triangle and undesired polymers, the stepwise method should be employed to design synthetic routes. The metallo-organic ligand **85**, containing four free tpy motifs for the final self-assembly, was prepared in seven steps and assembled with  $\text{Zn}^{2+}$  finally. Metallo-organic ligand **87** and hexatopic ligand **58** were assembled with  $\text{Cd}^{2+}$  to afford supramolecule **88**. With the same strategy, Wang group has reported metallo-supramolecular pentagram and hexagram with  $\text{Ru}^{2+}$  and  $\text{Cd}^{2+}$  or  $\text{Fe}^{2+}$ .<sup>[53]</sup>

To mimic nature's complex system and achieve the goal of increasing the complexity of terpyridinyl-based supramolecules, rhombus super snowflakes were designed and assembled by

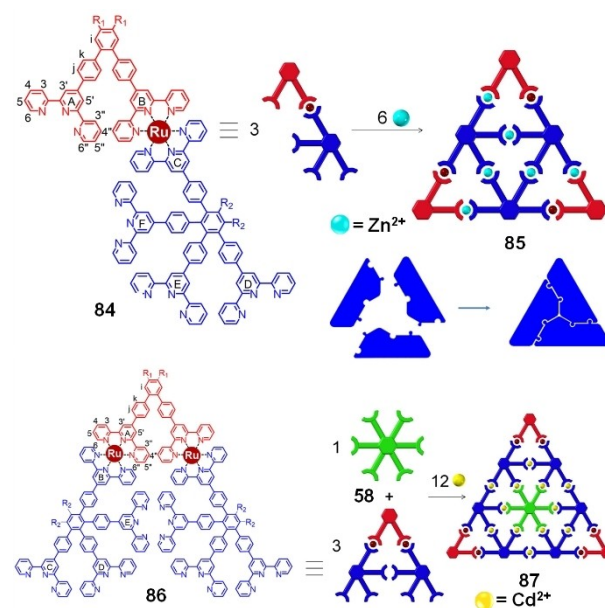


Figure 20. Synthetic routes of the supramolecular G2 Sierpiński triangle **85** and G3 Sierpiński triangle **87**.

the stepwise method (Figure 21).<sup>[54,55]</sup> When ligand **58** and V-typed and Y-typed ligands (1:6:6) in  $\text{CHCl}_3$  was added to a methanol solution of  $\text{Zn}(\text{NO}_3)_2$  (18 equiv) by one-step strategy, triangles were formed due to the outstanding self-sorting ability

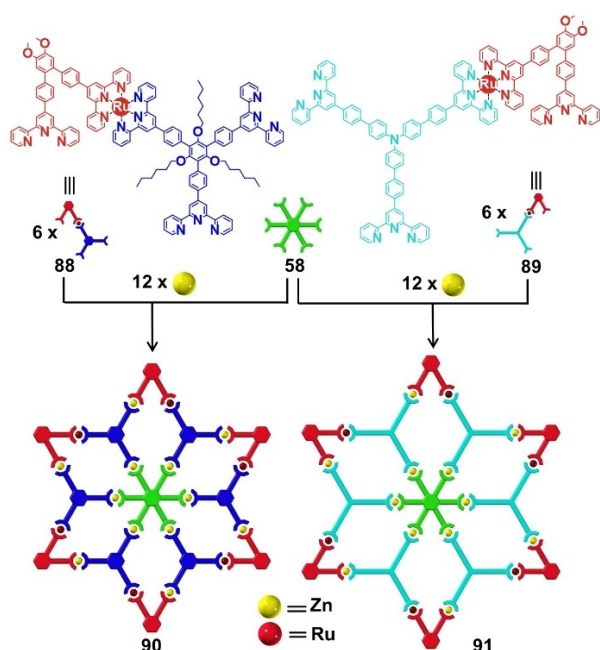


Figure 21. Self-assembly of snowflakes **90** and **91**.

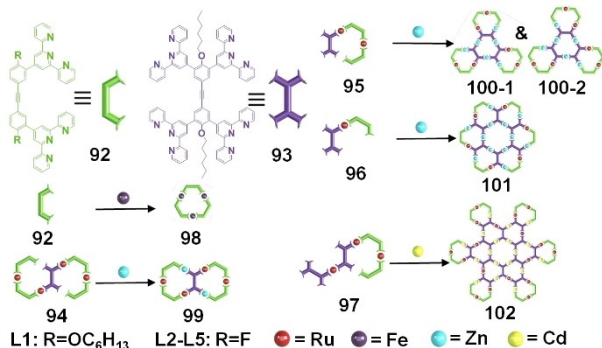


Figure 22. Self-assembly of fractal architectures **98**–**102**.

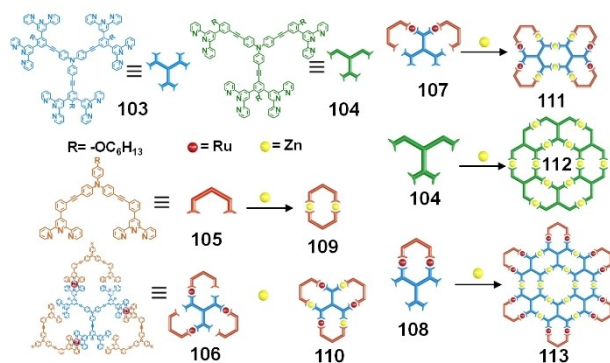


Figure 23. Self-assembly of fractal architectures **109**–**113**.

of V-typed ligand. To prevent the self-sorting ligand, metallo-ligand **88** was synthesized by using organic ligands and  $\text{Ru}^{2+}$ . Snowflake **90** was assembled with **88**, **58**, and  $\text{Zn}^{2+}$  in an exact ratio of 6:1:12, and the intermolecular dynamic exchange between **90** and **91** was studied and monitored by ESI-MS.

In the synthetic process, terpyridine motifs in multitopic ligands do not exhibit selectivity, and undesirable oligomers with a similar thermodynamic stability were obtained by the one-step method. It is crucial to construct fractal structures by a stepwise method. Li group has designed and constructed fractal structures with  $\text{Ru}^{2+}$  building blocks and  $\text{Zn}^{2+}$  or  $\text{Cd}^{2+}$  (Figure 22 and 23).<sup>[56,57]</sup> There are two ligands (i.e., **92** and **93**, **103** and **105**) in each system with similar angles to construct high-generation fractal supramolecules (**98**–**102**, **110**, **111**, and **113**) with high generation by linear building blocks (**93**–**97** and **106**–**108**) with specific sequences in two systems. With different sequences, the building blocks were obtained and further assembled with  $\text{Zn}^{2+}$  or  $\text{Cd}^{2+}$  to obtain well-defined structures. Structural details of supramolecules were verified by NMR and ESI-MS. The high-generation fractal supramolecules **101**, **102**, **112**, and **113** were hierarchically assembled into ordered nanofibers in solution or formed honeycomb networks on the highly oriented pyrolytic graphite (HOPG) surface based on the precisely-controlled shapes and sizes with high shape persistence.

Recently, Wang has constructed a series of elaborate molecular pinwheels **117**–**119** using preorganized metallo-organic ligands by the stepwise method (Figure 24).<sup>[58]</sup> In the design, the determining step is the delicate design of intricate metallo-organic ligands **114**–**116**, which combined multiple components (up to four) and  $\text{Ru}^{2+}$  in building blocks and synthesized with multi-fold Suzuki coupling reactions. The metallo-organic ligands were assembled with  $\text{Zn}^{2+}$  to afford the pinwheel-shaped star trigon **117** (16756 Da), pentagram **118** (28088.34 Da), and hexagram **119** (56053 Da), based on high DOCS.

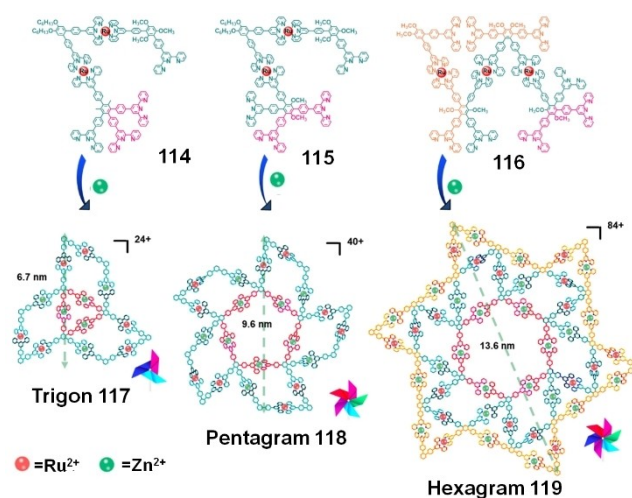
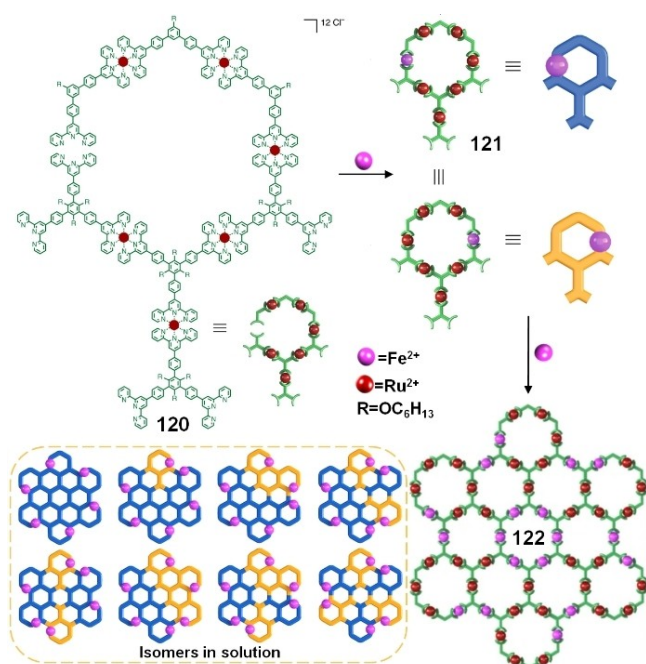
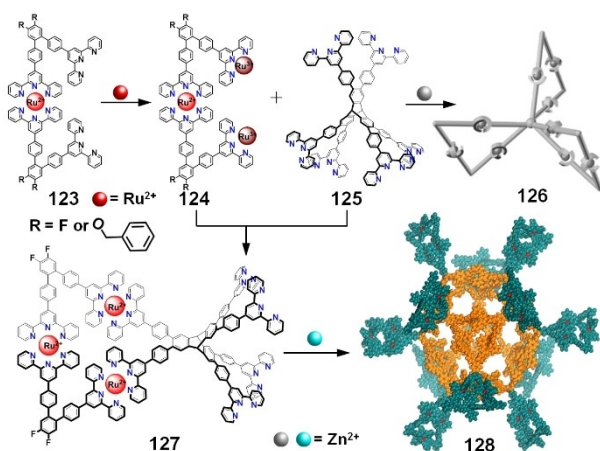


Figure 24. Self-assembly of pinwheel-shaped star trigon **117**, pentagram **118**, and hexagram **119**.

Recently, inspired by the precise and spontaneous self-assembly of proteins in biological organisms, Li and coauthors have reported the construction of a 20 nm-sized metallo-supramolecular hexagonal grid **122** in a stepwise manner,<sup>[59]</sup> which is the largest discrete 2D metallo-supramolecular architecture thus far (Figure 25). The metal-ligand **121** can flip over, leading to the different arrangements of the  $\text{Ru}^{2+}$  and  $\text{Fe}^{2+}$  centers, corresponding to isomers with unpredictable orders of metal ions existed in supramolecular system. Conventional solution-phase approaches such as NMR and ESI-MS were unable to distinguish isomers in a mixture. UHV-LT-STM was then utilized to further characterize grid **122** at the sub-molecule level. The order of metal sites was differentiated according to  $\text{Ru}^{2+}$  and  $\text{Fe}^{2+}$  due to different energy gaps by



**Figure 25.** Synthetic route of ligand **121** followed by the intra- and intermolecular self-assembly of supramolecular hexagonal grids **122** with  $\text{Fe}^{2+}$ .



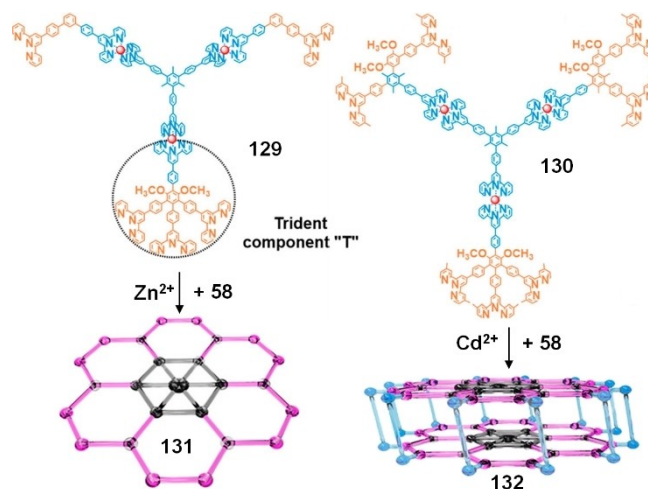
**Figure 26.** Self-assembly of spirane **126** and cuboctahedron **128**.

scanning tunnelling spectroscopy (STS). The possibility of occurrence for each isomer was counted by STS, which was in good agreement with the predicted based on theory. Each independent building block would not affect the formation of the supramolecule, and the main product with the smallest energy gap is the most optimal conformation. This study should advance our understanding of the design principles for constructing 2D metallo-supramolecules.

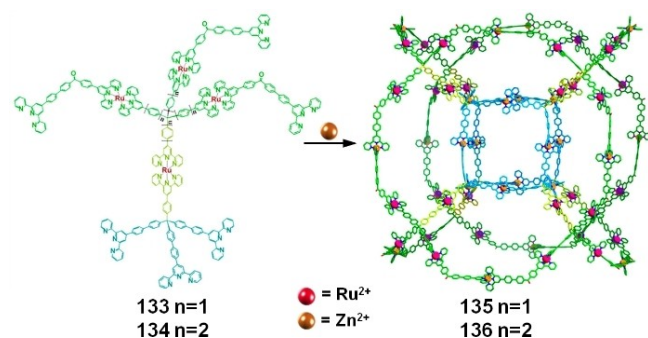
Except above 2D supramolecular structures, Newkome has reported the first 3D propeller-shaped spirane **126** with the tpy-Ru-tpy motif via a multi-component stepwise assembly (Figure 26).<sup>[60]</sup> First, the adduct **124** capped with  $\text{Ru}^{3+}$  was synthesized by metallo-dimer **123** with  $\text{RuCl}_3$ . As building blocks, hexaterypyridinyl triptycene **125** (1 equiv.) and adduct **124** (3 equiv.) were refluxed in a mixture of MeOH and  $\text{CHCl}_3$  with N-ethylmorpholine to afford spirane **126**. However, with different ratios of ligands, tetratopic heterotrimer **124** was assembled with intermediates **124** (1 equiv.) and hexaterypyridinyl triptycene **125** (1 equiv.). A supercharged (120+), large ( $d = 11.3$  nm) cuboctahedron **128**<sup>[61]</sup> was established with metallo-ligand **127** and  $\text{Zn}^{2+}$ . The supramolecular family was expanded by different ratios of ligands and metal ions.

By precise synthetic design, a 2D hexagon-fused wheel **131** and a double-decker prism **132** were assembled with metallo-organic ligands **129**, **130** and core hexatopic ligand **58** in a stepwise manner (Figure 27).<sup>[62]</sup> The prism structure **132** with 168 anions contained 36  $\text{Ru}^{2+}$  and 48  $\text{Cd}^{2+}$  with a molecular weight of up to 119 KDa, corresponding to a heteronuclear bimetallic supramolecular structure with the maximum charge. It is quite difficult to synthesize ligands by other methods. Wing-like metallo-organic ligands **129** and **130** with a trident component by tedious synthesis can perfectly connect with hexatopic ligand **58**. The introduction of hexatopic ligand **58** can considerably increase stability and complexity to the supramolecular architecture.

More recently, Wang designed and assembled two giant Russian doll-like molecular cubes **135** and **136** by increasing the



**Figure 27.** Self-assembly of hexagon-fused wheel **131** and prismatic lamellar structure **132**.



**Figure 28.** Synthesis of metallo-organic ligands 133 and 134 and energy-minimized structure of Russiandoll-like cubes 135 and 136.

coordination number by the stepwise method (Figure 28).<sup>[63]</sup> Different from a mixture assembled by the inner tritopic ligand, the predesigned hexatopic metalloligands 133 and 134 with geometric constraints were precisely assembled to well-defined double-shell cube architectures via the increase in the structural constraints. Cubes with a molecular weight of up to 75000 Da are well known to be the largest synthetic cage-in-cage supramolecules reported as yet.

## 5. Conclusion and outlook

In this review, an overview of the methods employed to design terpyridine-based metallo-supramolecules is provided. During the design and synthesis of supramolecular structures, it is crucial to select suitable methods for constructing predesigned architectures. Three self-assembly methods from simple systems to sophisticated systems were summarized in this Review: ligand geometry-directed, template-driven, and stepwise methods. In relatively simple self-assembly systems, the ligand geometry-directed method is extremely effective to afford highly ordered structures with thermodynamic stability. With the increasing complexity of systems, the ligand geometry-directed method and template-driven method are developed to avoid unwanted products and give predesigned structures. With the increase in supramolecular complexity, stepwise method is utilized to construct more complicated supramolecular structures by breaking the inherent self-assembly of ligands or unwanted product with disorganized manners. This Review could serve as guidance for the design of more complicated supramolecules in the future. Continuous efforts are anticipated in the field of supramolecules with aesthetic architectures via new methods, which in turn can promote the further pursuit of discrete artificial supramolecules.

## Conflict of Interest

The authors declare no conflict of interest.

**Keywords:** self-assembly method • discrete supramolecules • terpyridine

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